Robust Trust with Multiple Advisers

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Abstract

A decision-maker consults K advisers who observe the same evidence. Each adviser is independently aligned with probability α and otherwise may coordinate with all other misaligned advisers. Aligned advisers report the posterior belief truthfully; misaligned advisers observe the posterior and the entire alignment-type profile, but not the realized payoff state. We obtain two sharp results. First, with finite nondegenerate information and positive local payoff curvature, advice has positive worst-case value if and only if $\alpha > 1/2$, for every finite K and without symmetry of the posterior distribution. The impossibility proof supplies a common-fake jamming strategy and checks every report profile, including unanimous profiles. Second, under symmetric posteriors and quadratic loss, the exactly optimal aggregation rule is a margin ladder: a strict majority backed by net margin d is followed only inside a radius t_d determined by alignment odds $(\alpha/(1-\alpha))^d$; exact two-bloc ties are treated by the least-extreme rule, and other no-majority profiles are ignored. We prove optimality for arbitrary K by constructing a finite support saddle point channel by channel. Near $\alpha = 1/2$, K advisers have K times the single-adviser value. For fixed $\alpha > 1/2$, the remaining loss converges to zero at exponent $\mathrm{KL}(1/2 \,\|\, \alpha)$. Redundancy therefore amplifies already-positive trust but does not create trust below the one-half threshold in this common-evidence benchmark.

Keywords: robust design; multiple advisers; AI alignment; cheap talk; committees; worst-case aggregation.

1 Introduction

When several advisers analyze the same evidence, agreement seems reassuring and disagreement seems diagnostic. This intuition motivates second opinions, model ensembles, and proposals in

*I want to be explicit about the usage of the Artificial Intelligence tool. The unclaimed parts are done individually. I mainly used Codex for the following purposes: 1) Help me with the summary and understanding of the related literature; 2) Help me with converting from the preliminary drafts into formal LaTeX format; 3) Push forward the setting in the extension from 2 advisors to multiple agents. The way I used it was always to have a conversation with the advisors, such as "Could you help me summarize the idea of the third part of this paper?" "Can you tell me what the normal ways are of pushing this setting forward to multiple entities?" All the remaining errors are mine own.

which one AI system checks another. It is incomplete when adviser objectives are unknown. If misaligned advisers can coordinate and choose plausible reports strategically, the relevant question is not whether a majority is usually correct, but whether an aggregation rule still produces a guaranteed payoff improvement against the worst report strategy consistent with the model.

This paper solves that robust aggregation problem in a common-evidence benchmark. A binary payoff state generates a posterior belief μ. Every adviser observes μ and an independently drawn alignment type. An aligned adviser must report μ. A misaligned adviser observes μ and the full type profile and may coordinate with other misaligned advisers, but does not observe the realized state ω beyond the information contained in μ. The decision-maker commits to a rule before reports arrive and evaluates it by its worst-case expected payoff.

The first result identifies the trust threshold. For every finite number of advisers, advice has strictly positive robust value exactly when one adviser is more likely aligned than misaligned. The hard direction is distribution-free. For $\alpha \leq 1/2$, we pair each posterior above the prior with a random posterior below it using a mean-deviation transport. All advisers then report either the true posterior or the same transported fake, each coordinate choosing the two possibilities with equal unconditional probability. We verify that the posterior mean equals the prior after every mixed and unanimous on-path profile. This replaces the symmetry-only mirror argument and does not rely on a machine check. For $\alpha > 1/2$, following one adviser inside an arbitrarily small trust region yields a positive first-order gain when the value function has positive local curvature.

The second result gives the exact value for arbitrary K under symmetric finite posteriors and quadratic loss. Only a strict majority can move the action. If c of K advisers report the same value, its net margin is $d = 2c - K$. The report is clipped to a radius t_d around the prior, where

$$\left(\frac{\alpha}{1-\alpha}\right)^d \mathbb{E}[(\widetilde{m} - t_d)_+] = t_d + \mathbb{E}[\widetilde{m}], \qquad \widetilde{m} = |\mu - \mu_0|.$$

Thus consensus enters through log-odds: every additional net supporting adviser multiplies the truthful-to-fake channel odds by $\alpha/(1-\alpha)$. Conflicting numerical reports are not averaged. In an exact two-bloc tie the rule selects the less extreme same-side report, or the prior when reports straddle it. More fragmented profiles are ignored.

The optimality proof is an explicit saddle construction rather than an appeal to an unspecified "water-filling" adversary. For every margin d and every labeled majority set, we define (i) the drag demand of each exterior truthful report, (ii) the drag supply of each opposite-side source state, (iii) a product transportation plan matching those quantities, and (iv) the corresponding conditional report probabilities. Each attacked profile then has posterior mean exactly equal to the relevant clipping boundary. Interior profiles receive distinct off-support minority tags, and even-committee ties are paired with their reflected states. These channels are disjoint. The margin rule is therefore the Bayes response to the constructed adversary, while the same adversary attains the rule's pointwise worst loss for every state and type profile. The two inequalities meet and establish exact optimality among all deterministic, randomized, symmetric, or asymmetric rules.

The redundancy results follow from the exact formula. Close to the threshold, the value is quadratic in $\alpha - 1/2$ and scales exactly by K to first order. With fixed quality above the threshold, the loss relative to full information has the large-deviation exponent of an aligned committee failing to form a strict majority. A local quality-equivalent calculation implies that two advisers of quality $1/2 + \varepsilon$ are worth one adviser of quality $1/2 + \sqrt{2}\varepsilon + o(\varepsilon)$. We deliberately do not extrapolate this local equivalence into an unproved global dominance claim.

Relation to the literature. The model extends the single-adviser robust-trust problem of Dworczak and Smolin (2026); the common posterior, rich action environment, and trust-region logic come from that framework. The new objects are multiple unknown types, ordinal disagreement handling, margin-specific trust regions, and an arbitrary-K saddle construction. Fudenberg and Liang (2025) instead restrict what a possibly misaligned AI can learn or report using population information; this is an input-side instrument and can break the impersonation channel that drives our output-side threshold. Alonso, Gan, and Hu (2026) study robust delegation under preference uncertainty, a related max–min design problem with a different delegation-set technology. Our full-adversary benchmark also differs from classical strategic communication (Crawford and Sobel, 1982) and multi-sender models with known preferences (Battaglini, 2002; Krishna and Morgan, 2001; Ambrus and Takahashi, 2008). The common signal deliberately removes the Blackwell information gain from an independent second signal (Blackwell, 1951), isolating verification value. Debate experiments motivate the application but do not identify the robust optimum studied here (Irving, Christiano, and Amodei, 2018; Khan et al., 2024).

2 Model and scope

There is a binary state $\omega \in \{0, 1\}$ with prior $\mu_0 = \mathbb{P}(\omega = 1)$. A common signal induces the posterior belief

$$\mu = \mathbb{P}(\omega = 1 \mid s), \qquad \mu \sim \tau, \qquad \mathbb{E}_\tau[\mu] = \mu_0.$$

The support $S \subset [0, 1]$ of τ is finite. The equality of the mean and the prior is Bayes plausibility. There are $K \geq 1$ advisers. Their types $\theta_i \in \{A, M\}$ are independent of (ω, μ) and of one another, with $\mathbb{P}(\theta_i = A) = \alpha$. All advisers observe $(\mu, \theta_1, \ldots, \theta_K)$, but not ω separately. Adviser i must report $m_i = \mu$ when $\theta_i = A$. Misaligned advisers choose an arbitrary joint distribution of their messages, conditional on $(\mu, \theta_1, ldots, \theta_K)$, and may use shared and private randomization.

The decision-maker chooses an action indexed by a belief $\nu \in [0, 1]$. Its expected payoff at true belief μ is

$$W(\nu, \mu) = U(\nu) + U'(\nu)(\mu - \nu), \tag{1}$$

where U is convex. Acting on the true belief gives $U(\mu)$. A rule a maps report profiles to distributions over actions. Its robust value relative to the prior action is

$$V_K = \sup_a \inf_\beta \mathbb{E}_{\tau,\theta,\beta,a}[W(a(\mathbf{m}),\mu) - W(\mu_0,\mu)]\,, \tag{2}$$

where β ranges over all admissible coordinated misaligned strategies.

We use two explicit assumption sets.

Assumption 1 (Threshold environment)**.** *τ is finite and nondegenerate, and $U \in C^2$ satisfies $U''(\mu_0) > 0$. No symmetry of τ is imposed.*

Assumption 2 (Exact-value environment)**.** *In addition to finite nondegenerate support, τ is symmetric about μ_0: $\tau(u) = \tau(2\mu_0 - u)$ whenever $u \in S$. Payoff is quadratic, so regret from action ν at belief μ is $(\nu - \mu)^2$.*

Assumption 1 states the local curvature actually used in the positivity proof. Strict convexity and C^2 alone would not imply $U''(\mu_0) > 0$ (for example, a fourth-power function is strictly convex but flat to second order at zero). Assumption 2 is the boundary of the closed-form margin theorem; the asymmetric exact aggregation problem is not claimed as solved here.

Lemma 1 (Posterior-mean reduction)**.** *Fix any adversary β. At every on-path profile $\mathbf{m}$, an optimal action is $\nu = \mathbb{E}_\beta[\mu \mid \mathbf{m}]$. Under quadratic loss the resulting improvement over the prior is*

$$I(\beta) = \mathrm{Var}_\beta(\mathbb{E}_\beta[\mu \mid \mathbf{m}])\,.$$

Consequently $V_K \le \inf_\beta I(\beta)$.

The proof is in Appendix A.1. The information inequality is used only for upper bounds; exactness later comes from a rule and adversary that form a saddle point.

3 The trust threshold

Theorem 1 (Trust threshold for arbitrary finite posteriors)**.** *Under Assumption 1, for every finite $K \ge 1$,*

$$V_K > 0 \quad \Longleftrightarrow \quad \alpha > \frac{1}{2}.$$

For $\alpha \le 1/2$ the same zero-value conclusion holds for every convex differentiable U for which the rich action representation (1) is well defined.

The proof in Appendix A.2 is constructive. Its key accounting number is

$$S_\tau = \sum_{u>\mu_0} \tau(u)(u - \mu_0) = \sum_{v<\mu_0} \tau(v)(\mu_0 - v) > 0. \tag{3}$$

The equality follows from $\mathbb{E}[\mu] = \mu_0$. A high truth draws a common low fake in proportion to the low state's contribution to the right side of (3); a low truth does the opposite. Misaligned advisers mix between the truth and this common fake so that each coordinate is equally likely to display either. Pairwise posterior drags cancel. The appendix separately checks two-valued profiles, unanimous profiles, and a possible atom at the prior.

Remark 1 (What is and is not universal)**.** *The one-half threshold is invariant to K and to asymmetry of the finite posterior distribution in this common-evidence, independently typed, rich-action model. It is not a claim about independent signals, correlated adviser types, state-informed saboteurs, constrained messages, or coarse action sets. Those changes remove steps used by the jamming construction and may change the threshold.*

4 Exact aggregation under symmetry and quadratic loss

Maintain Assumption 2 and $\alpha > 1/2$. Write

$$\widetilde{m} = |\mu - \mu_0|, \qquad M = \mathbb{E}[\widetilde{m}], \qquad \bar{m} = \max \widetilde{m},$$

and define the truthful gain and adversarial damage from radius t by

$$G(t) = \mathbb{E}\Big[\widetilde{m}^2 - (\widetilde{m} - t)_+^2\Big], \tag{4}$$

$$D(t) = \mathbb{E}[(\widetilde{m} + t)^2 - \widetilde{m}^2] = 2Mt + t^2. \tag{5}$$

For every positive integer margin d, let t_d be the unique solution in $(0, \bar{m})$ to

$$r^d \mathbb{E}[(\widetilde{m} - t_d)_+] = t_d + M, \qquad r = \frac{\alpha}{1 - \alpha} > 1. \tag{6}$$

Appendix A.3 proves existence, uniqueness, and $t_{d+1} > t_d$.

4.1 The margin rule

For a report profile $\mathbf{m}$, count identical report values.

(i) If a value u appears $c > K/2$ times, it is the unique strict majority. Put $d = 2c - K$ and choose $a(\mathbf{m}) = \text{clip}(u; \mu_0 - t_d, \mu_0 + t_d)$.

(ii) If K is even and exactly two values each appear $K/2$ times, choose μ_0 when their closed interval contains μ_0. If both lie on the same side, choose the one closer to μ_0 and clip it to $[\mu_0 - t_K, \mu_0 + t_K]$.

(iii) In every other profile, choose μ_0.

The word "majority" is essential. A plurality of two reports in a 2–2–1 profile does not move the action. Only equality of report values matters; the rule never averages disagreeing numbers.

Let

$$w_{K,j} = \binom{K}{j} \alpha^{K-j}(1-\alpha)^j \tag{7}$$

be the probability that exactly j advisers are misaligned.

Theorem 2 (Margin-ladder value and optimal rule)**.** *Under Assumption 2, for every finite K and $\alpha > 1/2$,*

$$V_K = \sum_{j<K/2} w_{K,j} G(t_{K-2j}) - \sum_{j>K/2} w_{K,j} D(t_{2j-K}). \tag{8}$$

The margin rule attains this value and is optimal among all rules, including randomized and asymmetric rules. There exists an explicit coordinated adversary β^* such that the margin rule is a Bayes response to β^* and β^* attains the rule's worst loss separately for every posterior state and alignment-type profile.

Appendices A.4–A.5 give the full proof. The construction is valid for arbitrary K; no step is delegated to a $K \leq 3$ machine check. The numerical program is an independent audit of the algebra, not part of the proof.

Corollary 1 (Two advisers)**.** *For $K = 2$, let t_2 solve $r^2\mathbb{E}[(\widetilde{m} - t_2)_+] = t_2 + M$. Then*

$$V_2 = \alpha^2 G(t_2) - (1-\alpha)^2 D(t_2) \tag{9}$$

$$= \alpha^2 \mathbb{E}[\min\{\widetilde{m}, t_2\}^2] + (1-\alpha)^2 t_2^2. \tag{10}$$

On disagreement, the optimal rule acts on the prior when reports straddle it and otherwise follows the less extreme report, clipped at t_2. In particular, a numerical average is not robustly optimal.

Equation (10) follows from (9) and the radius budget; the algebra is shown explicitly in Appendix A.6.

5 The value of redundancy

Corollary 2 (Near-threshold scaling, convergence, and local quality equivalent)**.** *Under Assumption 2:*

(i) *As $\alpha \downarrow 1/2$,*

$$\frac{V_K(\alpha)}{V_1(\alpha)} \longrightarrow K.$$

(ii) *For every fixed $\alpha > 1/2$,*

$$-\lim_{K\to\infty} \frac{1}{K} \log\left(1 - \frac{V_K}{\mathrm{Var}(\mu)}\right) = \mathrm{KL}\left(\frac{1}{2}\middle\|\alpha\right).$$

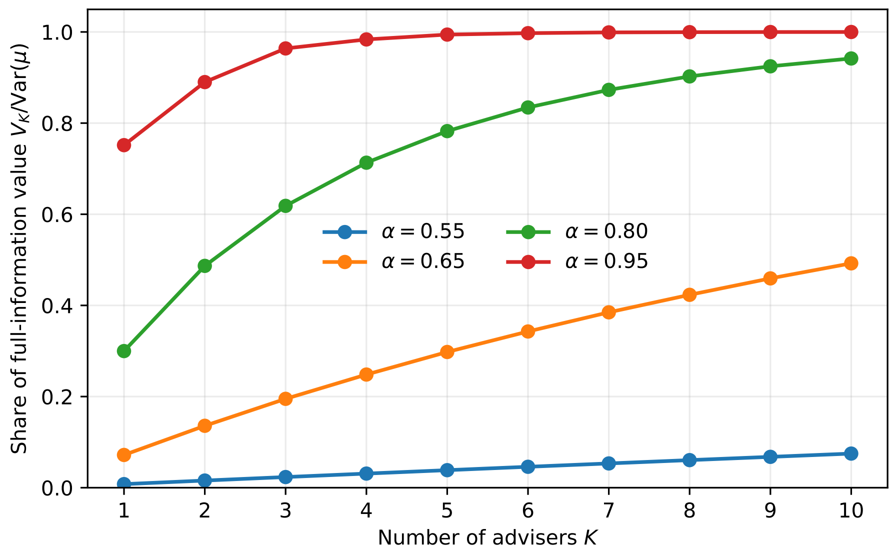

Figure 1: Verified values for the uniform nine-point posterior distribution. The vertical scale is the share of full-information value. Curves are generated by the closed-form formula, not by a fitted simulation.

(iii) *Let $f(\alpha)$ be the local solution above $1/2$ of $V_1(f(\alpha)) = V_2(\alpha)$. Then*

$$f(\alpha) = \frac{1}{2} + \sqrt{2}\left(\alpha - \frac{1}{2}\right) + o\left(\alpha - \frac{1}{2}\right).$$

Appendix A.7 derives all constants. If $p_0 = \mathbb{P}(\widetilde{m} > 0)$, then for sufficiently small t, $G(t) = 2Mt - p_0 t^2$ and $D(t) = 2Mt + t^2$. This gives

$$V_K\left(\frac{1}{2} + \varepsilon\right) = \frac{8KM^2}{1 + p_0}\varepsilon^2 + o(\varepsilon^2), \tag{11}$$

which proves both (i) and (iii). Part (ii) combines a binomial lower bound with a finite-support bound $\bar{m} - t_d \le Cr^{-d}$ on good-majority clipping loss.

6 Interpretation, limitations, and extensions

Output-side redundancy is threshold-rigid. The result says that a committee can magnify value once individual quality exceeds one half, but it cannot bootstrap positive robust value from individually sub-threshold advisers in the stated benchmark. This is an output-side statement: it holds while misaligned messages remain unrestricted and every adviser sees the same posterior.

Input-side verification can change the problem. Audits, report admissibility, externally checkable statistics, and independently generated evidence can remove fake channels used in the

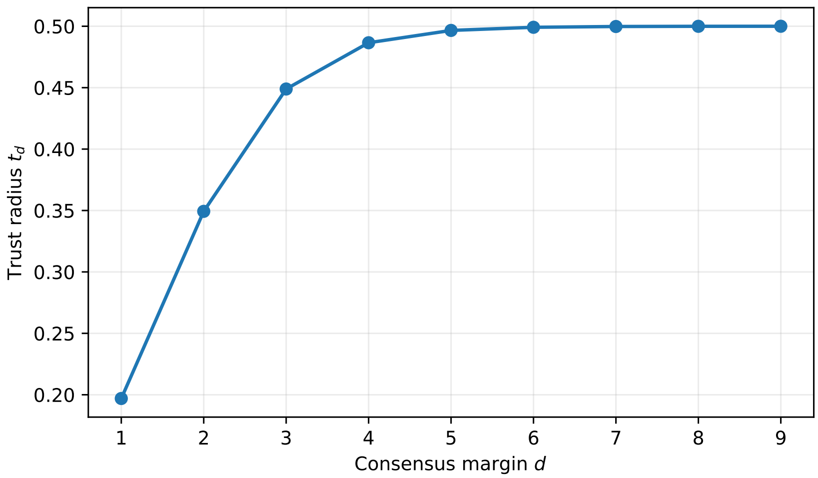

Figure 2: Trust radius by consensus margin at $\alpha = .80$ in the same environment. A larger net majority raises the truthful-to-fake odds and therefore widens the trust region.

proof. The framework of Fudenberg and Liang (2025) illustrates how population restrictions discipline a possibly misaligned AI. It would be incorrect to infer from Theorem 1 that no form of cross-checking can lower a trust threshold; the theorem concerns cross-checking unrestricted reports about common evidence.

Asymmetry. The threshold theorem already permits asymmetric τ. The exact margin ladder does not. Under asymmetry one should generally expect different upper and lower trust radii and coupled transport budgets. Version 2 stated this extension too strongly on numerical evidence. Version 3 treats a closed-form asymmetric saddle point as an open extension until a complete analytic proof is available.

Other boundaries. Correlated types change the likelihood ratio attached to a margin. Independent adviser signals add Blackwell information value to the verification value isolated here. A saboteur who observes ω rather than only μ can condition messages on payoff-relevant information absent from the formal model and therefore defines a different game. Coarse actions can also invalidate the local-curvature positivity argument. These are substantive extensions, not innocuous changes of notation.

7 Reproducibility and claim status

The analytical proof is self-contained. The deterministic script `replication/code/verify_results.py` performs three audits on a uniform nine-point posterior support: it solves every radius equation and evaluates (8); it implements the explicit saddle distribution and checks every on-path posterior and every state–type conditional loss; and, for $K \leq 3$, it solves a separate finite-action epigraph linear program. The standard run checks $\alpha \in \{.55, .65, .80, .95\}$ and $K \leq 6$. Across these 24 cases the largest probability-row error is below 9×10^{-13}, the largest posterior-action or pointwise-loss error is below 2.3×10^{-16}, and the largest aggregate formula error is below 1.1×10^{-14}; the independent LP differs from the formula by at most 1.4×10^{-16}. Results are stored in `replication/results/verification_summary.json` and `replication/results/verification_table.csv`. These calculations audit implementation and algebra; they are not substituted for the arbitrary-K proof.

A Detailed proofs

This appendix supplies every probability kernel and every equality used in the main results. Empty sums below are zero. Because S is finite, conditional distributions can be defined pointwise and all expectations are finite sums.

A.1 Posterior-mean reduction

Proof of Lemma 1. Fix an adversary β and an on-path report profile $\mathbf{m}$. Let $\widehat{\mu}(\mathbf{m}) = \mathbb{E}_\beta[\mu \mid \mathbf{m}]$. If the decision-maker chooses the action indexed by ν, conditional expected payoff is

$$\begin{aligned}
\mathbb{E}_\beta[W(\nu, \mu) \mid \mathbf{m}] &= \mathbb{E}_\beta[U(\nu) + U'(\nu)(\mu - \nu) \mid \mathbf{m}] \\
&= U(\nu) + U'(\nu)(\widehat{\mu}(\mathbf{m}) - \nu) \\
&= W(\nu, \widehat{\mu}(\mathbf{m})).
\end{aligned}$$

By the definition of the rich action representation, this is maximized at $\nu = \widehat{\mu}(\mathbf{m})$ and the maximized value is $U(\widehat{\mu}(\mathbf{m}))$. Randomizing over actions cannot improve the maximum because expected payoff is the average of the payoffs of the actions in the mixture.

Under quadratic loss, the expected loss of the posterior-mean action is

$$\begin{aligned}
\mathbb{E}[(\mu - \widehat{\mu})^2] &= \mathbb{E}[\mathrm{Var}(\mu \mid \mathbf{m})] \\
&= \mathrm{Var}(\mu) - \mathrm{Var}(\mathbb{E}[\mu \mid \mathbf{m}]),
\end{aligned}$$

where the last equality is the law of total variance. The prior action has loss $\mathrm{Var}(\mu)$, so the improvement is $I(\beta) = \mathrm{Var}(\mathbb{E}[\mu \mid \mathbf{m}])$.

Finally, the decision-maker's payoff against a particular admissible adversary cannot exceed her Bayes payoff against that adversary. Therefore, for every β, $V_K \leq I(\beta)$, and taking the infimum gives $V_K \leq \inf_\beta I(\beta)$. □

A.2 The arbitrary-prior jamming construction

Proof of Theorem 1. We prove zero value for $\alpha \leq 1/2$ and positive value for $\alpha > 1/2$ separately.

Step 1: define an opposite-side fake kernel. Let S_τ be defined by (3). Nondegeneracy and Bayes plausibility imply $S_\tau > 0$. Define a kernel $Q(\cdot \mid u)$ as follows:

$$Q(v \mid u) = \begin{cases} \dfrac{\tau(v)(\mu_0 - v)}{S_\tau}, & u > \mu_0,\ v < \mu_0, \\ \dfrac{\tau(v)(v - \mu_0)}{S_\tau}, & u < \mu_0,\ v > \mu_0, \\ 1, & u = v = \mu_0, \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

For $u > \mu_0$, summing the first line over $v < \mu_0$ gives one by (3); the same is true below the prior. Thus Q is a probability kernel.

For every $u > \mu_0 > v$, the two directed pairs satisfy the key identity

$$\begin{aligned} &\tau(u)Q(v \mid u)(u - \mu_0) + \tau(v)Q(u \mid v)(v - \mu_0) \\ &\quad = \frac{\tau(u)\tau(v)(u - \mu_0)(\mu_0 - v)}{S_\tau} - \frac{\tau(v)\tau(u)(\mu_0 - v)(u - \mu_0)}{S_\tau} = 0. \end{aligned} \tag{13}$$

Step 2: define the advisers' reports. At truth $\mu = u \neq \mu_0$, draw one common fake Z from $Q(\cdot \mid u)$. Conditional on (u, Z), every aligned adviser reports u. Every misaligned adviser independently reports u with probability

$$\gamma = \frac{1 - 2\alpha}{2(1 - \alpha)} \tag{14}$$

and reports the common Z otherwise. When $u = \mu_0$, all advisers report μ_0. For $\alpha \leq 1/2$, $\gamma \in [0, 1/2]$, so this is feasible. For any one coordinate and fixed (u, Z),

$$\begin{aligned} \mathbb{P}(m_i = u \mid u, Z) &= \alpha + (1 - \alpha)\gamma = \frac{1}{2}, \\ \mathbb{P}(m_i = Z \mid u, Z) &= (1 - \alpha)(1 - \gamma) = \frac{1}{2}. \end{aligned}$$

The type draws and the misaligned advisers' private coins are independent across coordinates. Hence, conditional on (u, Z), every particular ordered vector in $\{u, Z\}^K$ has probability 2^{-K}.

Step 3: check a profile containing two distinct values. Consider an ordered profile $\mathbf{m}$ whose distinct values are $u > \mu_0 > v$. Suppose a specified set of coordinates reports u and the complement reports v. There are exactly two state–fake pairs that generate this profile: $(\mu, Z) = (u, v)$ and $(\mu, Z) = (v, u)$. The respective unconditional masses are

$$2^{-K}\tau(u)Q(v \mid u) \quad \text{and} \quad 2^{-K}\tau(v)Q(u \mid v).$$

The numerator of the posterior displacement from μ_0 is therefore

$$2^{-K}\{\tau(u)Q(v \mid u)(u-\mu_0) \\ +\tau(v)Q(u \mid v)(v-\mu_0)\} = 0$$

by (13). Thus $\mathbb{E}[\mu \mid \mathbf{m}] = \mu_0$.

Step 4: check a unanimous non-prior profile. Let every coordinate report $u > \mu_0$. This can occur in two ways. First, the truth is u, the fake is any $v < \mu_0$, and all coordinates select the truth. Second, the truth is any $v < \mu_0$, the common fake is u, and all coordinates select the fake. The posterior-displacement numerator, after removing the common factor 2^{-K}, is

$$\begin{aligned}&\tau(u)(u-\mu_0)\sum_{v<\mu_0} Q(v \mid u) + \sum_{v<\mu_0} \tau(v)Q(u \mid v)(v-\mu_0)\\ &= \sum_{v<\mu_0} [\tau(u)Q(v \mid u)(u-\mu_0) + \tau(v)Q(u \mid v)(v-\mu_0)] = 0.\end{aligned}$$

The first equality uses that $Q(\cdot \mid u)$ sums to one; the second is (13) summed over v. The case $u < \mu_0$ is symmetric. A unanimous report of μ_0 can arise only when the truth is μ_0, so its posterior mean is also μ_0.

Step 5: conclude zero value. Steps 3 and 4 exhaust the on-path profiles: the strategy uses at most the truth and one common fake. Every on-path posterior mean equals μ_0. Lemma 1 implies that even a decision-maker who knows this fixed adversary cannot outperform the prior action. The prior action itself guarantees zero relative value against every adversary. Hence $V_K = 0$ for $\alpha \le 1/2$. Notice that the argument did not use symmetry of τ, quadratic loss, or coordination across fake choices beyond drawing the common Z.

Step 6: construct a positive-value rule above one half. Suppose $\alpha > 1/2$. Ignore advisers $2, \dots, K$. For $t > 0$, map adviser 1's report to

$$a_t(m_1) = \operatorname{clip}(m_1; \mu_0 - t, \mu_0 + t).$$

Because S is finite, choose t smaller than $\min\{|u-\mu_0| : u \in S,\ u \neq \mu_0\}$. For any non-prior u, write $s_u = \operatorname{sign}(u-\mu_0)$.

Differentiate (1) with respect to the action index:

$$\frac{\partial}{\partial \nu} W(\nu, u) = U''(\nu)(u-\nu). \tag{15}$$

If adviser 1 is aligned, its induced action is $\mu_0 + s_u t$. By (15) and a uniform Taylor expansion over the finite support,

$$W(\mu_0 + s_u t, u) - W(\mu_0, u) = \int_0^t U''(\mu_0 + s_u x)(|u - \mu_0| - x)\, dx$$
$$= U''(\mu_0)|u - \mu_0|t + o(t). \tag{16}$$

If adviser 1 is misaligned, every reachable action lies in $[\mu_0 - t, \mu_0 + t]$. The worst endpoint to first order is $\mu_0 - s_u t$, and the adversary can reach it by reporting that endpoint. Again by (15),

$$\min_{\nu \in [\mu_0 - t, \mu_0 + t]} \{W(\nu, u) - W(\mu_0, u)\} = -U''(\mu_0)|u - \mu_0|t + o(t). \tag{17}$$

The atom $u = \mu_0$ contributes only $O(t^2)$. Averaging (16)–(17) gives the guaranteed improvement

$$(2\alpha - 1)U''(\mu_0)\mathbb{E}|\mu - \mu_0|\, t + o(t).$$

All three leading factors are strictly positive. The expression is therefore positive for all sufficiently small $t > 0$. Thus $V_1 > 0$, and ignoring $K - 1$ advisers implies $V_K \geq V_1 > 0$. □

A.3 Trust radii

Lemma 2 (Radius existence and monotonicity)**.** *For $r > 1$ and each integer $d \geq 1$, equation (6) has exactly one solution $t_d \in (0, \bar{m})$. If $d' > d$, then $t_{d'} > t_d$.*

Proof. Define

$$F_d(t) = r^d \mathbb{E}[(\widetilde{m} - t)_+] - (t + M), \qquad t \in [0, \bar{m}].$$

It is continuous. Since $M > 0$, $F_d(0) = (r^d - 1)M > 0$, while $F_d(\bar{m}) = -(\bar{m} + M) < 0$. Moreover, if $t' > t$, then $(\widetilde{m} - t')_+ \leq (\widetilde{m} - t)_+$ state by state and $t' + M > t + M$; hence $F_d(t') < F_d(t)$. The intermediate value theorem and strict decrease give a unique root in the open interval.

At $t = t_d$, the expectation $\mathbb{E}[(\widetilde{m} - t_d)_+]$ is positive because the right side of (6) is positive. Therefore

$$F_{d'}(t_d) = (r^{d'} - r^d)\mathbb{E}[(\widetilde{m} - t_d)_+] > 0.$$

Because $F_{d'}$ is strictly decreasing and vanishes at $t_{d'}$, its root must satisfy $t_{d'} > t_d$. □

A.4 The margin rule's pointwise worst case

Fix a truth μ and a type profile with exactly j misaligned advisers. Put $m = |\mu - \mu_0|$.

Proposition 1 (Conditional worst loss)**.** *The supremum of the margin rule's conditional quadratic* *loss over the misaligned reports equals*

$$L_j(\mu) = \begin{cases} (m - t_{K-2j})_+^2, & j < K/2, \\ m^2, & j = K/2, \\ (m + t_{2j-K})^2, & j > K/2. \end{cases} \tag{18}$$

The middle line is relevant only when K is even.

Proof. If $j < K/2$, the $K - j$ aligned advisers all report μ and form a strict majority, irrespective of the misaligned messages. Its smallest possible margin occurs when none of the j misaligned advisers joins the truth and equals $2(K - j) - K = K - 2j$. If some join the truth, the margin and hence the trust radius weakly increase by Lemma 2; if the others split across several fake values, the truthful count does not change. Thus the action is the truth clipped at a radius of at least t_{K-2j}. It lies on the segment from μ_0 to μ, so loss is at most $(m - t_{K-2j})_+^2$. The adversary attains equality by assigning all j fake reports a common value different from μ, leaving the truthful margin exactly $K - 2j$.

If $j > K/2$, the misaligned advisers can report a common value at the endpoint opposite the truth. They then form a strict majority of margin $2j - K$ and induce the action $\mu_0 - t_{2j-K}$ when $\mu > \mu_0$, or $\mu_0 + t_{2j-K}$ when $\mu < \mu_0$. The resulting loss is $(m + t_{2j-K})^2$. No report pattern can do worse: a fake strict majority with $c \leq j$ has margin $2c - K \leq 2j - K$ and hence reaches no farther from the prior; a no-majority profile induces the prior; and a truthful majority induces an action between the prior and the truth.

If $j = K/2$, aligned advisers form one bloc at μ. If the fake bloc reports the opposite side, the rule selects the prior. If it reports the same side, the rule selects the less extreme of the two bloc values, clipped if necessary. In either case the action lies on the closed segment between μ_0 and μ, so loss is at most m^2. Reporting the reflected value $2\mu_0 - \mu$ makes the reports straddle the prior and attains m^2. $\square$

Averaging (18) with the weights (7), and subtracting from the prior loss $\mathbb{E}[m^2] = \mathrm{Var}(\mu)$, gives exactly the right side of (8). Indeed, a good-majority event improves loss by $G(t)$, a bad-majority event worsens it by $D(t)$, and a tie makes no change. This proves that the displayed value is a lower bound on V_K.

A.5 Explicit saddle construction for arbitrary K

We now construct one adversary that (i) attains every conditional loss in Proposition 1 and (ii) makes the margin rule a posterior-mean action at every on-path profile. Together these properties prove the matching upper bound.

Fix a feasible margin $d \in \{1, \ldots, K\}$ with $d \equiv K \pmod 2$. Define

$$c = \frac{K + d}{2}, \qquad \ell = \frac{K - d}{2}, \qquad p_d^+ = \alpha^c (1 - \alpha)^\ell, \qquad p_d^- = \alpha^\ell (1 - \alpha)^c.$$

These are probabilities of two *labeled* complementary type profiles; binomial coefficients will arise only when we sum over the possible labeled majority sets. Their ratio is $p_d^+/p_d^- = r^d$.

Step 1: define high-side drag demand and supply. Let $b_d^+ = \mu_0 + t_d$. For every exterior high state $u > b_d^+$, define demand

$$D_d^+(u) = p_d^+ \tau(u)(u - b_d^+). \tag{19}$$

For every low source $v < \mu_0$, define supply

$$S_d^+(v) = p_d^- \tau(v)(b_d^+ - v). \tag{20}$$

If τ has an atom at μ_0, assign half of that atom to the high-side source pool, with $S_d^+(\mu_0) = \frac{1}{2} p_d^- \tau(\mu_0) t_d$. Symmetry gives

$$\sum_{u > b_d^+} D_d^+(u) = \frac{p_d^+}{2} \mathbb{E}[(\widetilde{m} - t_d)_+], \tag{21}$$

$$\sum_{v \le \mu_0} S_d^+(v) = \frac{p_d^-}{2}(t_d + M). \tag{22}$$

The radius equation and $p_d^+/p_d^- = r^d$ make (21) and (22) equal. Denote their positive common value by T_d^+.

Step 2: transport drag and convert it into probabilities. For every high target u and low-or-prior source v in the preceding pools, define the product transport

$$z_d^+(v, u) = \frac{S_d^+(v) D_d^+(u)}{T_d^+}. \tag{23}$$

Its row sums are $S_d^+(v)$ and its column sums are $D_d^+(u)$. Convert drag into conditional report probabilities:

$$\sigma_d^+(v \mid u) = \frac{z_d^+(v, u)}{D_d^+(u)}, \qquad \kappa_d^+(u \mid v) = \frac{z_d^+(v, u)}{p_d^- \tau(v)(b_d^+ - v)}. \tag{24}$$

Thus $\sum_v \sigma_d^+(v \mid u) = 1$. For $v < \mu_0$, the denominator in κ_d^+ equals $S_d^+(v)$ and its row sums to one. For $v = \mu_0$, the denominator is twice $S_d^+(\mu_0)$ and the high-side row sums to one half. The low-side construction supplies the other half. The two half-rows together, rather than either side separately, form the prior state's conditional report distribution.

The symmetric low-side construction uses boundary $b_d^- = \mu_0 - t_d$, exterior targets $u < b_d^-$, sources $v > \mu_0$, and the other half of a prior atom. Denote the resulting objects by $D_d^-, S_d^-, z_d^-, \sigma_d^-, \kappa_d^-$. The same calculation proves feasibility.

Step 3: specify reports for each labeled type profile. Fix a subset $B \subseteq \{1, \ldots, K\}$.

If $|B| = c > K/2$, interpret B as the aligned set. At an exterior truth $u > b_d^+$, advisers in B report u, while the ℓ misaligned advisers draw one common v from $\sigma_d^+(\cdot \mid u)$ and report v. At an exterior low truth use σ_d^-. At an interior truth $|u - \mu_0| \le t_d$, the misaligned minority reports a common tag $\eta_{B,u}$ chosen outside S. Tags are distinct across (B, u) and across channels. Such tags exist because S and the collection of channels are finite while the message interval is infinite. When $\ell = 0$, no tag is needed and the profile is unanimously truthful.

If $|B| = c > K/2$, we also need the complementary type profile in which B is the misaligned set and B^c is aligned. At a low-or-prior truth v, all advisers in B draw a common exterior high target u from $\kappa_d^+(\cdot \mid v)$ and report u; advisers in B^c report the truth v. Use κ_d^- at high-or-prior truths. At a prior atom, use the union of the high and low rows; each has row sum one half by Step 2. This exhausts every bad-majority conditional row.

For even K, when exactly $K/2$ advisers are aligned, every misaligned adviser reports the reflected truth $\rho(u) = 2\mu_0 - u$. Finally, all rules just described are applied separately to every labeled aligned set. This defines β^* for every state and type profile.

Step 4: compute the posterior at an attacked high profile. Fix B of size c, a high target $u > b_d^+$, and a source $v \le \mu_0$. Consider the labeled profile in which advisers in B report u and advisers in B^c report v. By the use of unique tags and disjoint target sets, exactly two constructed events produce this profile:

(a) the truth is u, B is aligned, and the minority draws v; or

(b) the truth is v, B^c is aligned, and the fake majority draws u.

Their masses are

$$q_u = p_d^+ \tau(u) \sigma_d^+(v \mid u) = \frac{z_d^+(v, u)}{u - b_d^+},$$
$$q_v = p_d^- \tau(v) \kappa_d^+(u \mid v) = \frac{z_d^+(v, u)}{b_d^+ - v}.$$

The second equalities substitute (19)–(24). Therefore

$$q_u(u - b_d^+) = z_d^+(v, u) = q_v(b_d^+ - v),$$

or, equivalently,

$$\frac{q_u u + q_v v}{q_u + q_v} = b_d^+ = \mu_0 + t_d.$$

The posterior mean at this profile is exactly the high clipping boundary selected by the margin rule. The low-side calculation gives posterior mean $b_d^- = \mu_0 - t_d$.

Step 5: compute posteriors at every remaining profile. An interior truthful-majority profile contains its unique tag and hence identifies its truth u; its posterior mean is u, which lies inside the margin-d trust region. A unanimous exterior profile is covered by Step 4 with $\ell = 0$: it pools an all-aligned exterior truth with all-misaligned opposite sources and has posterior at the boundary. A unanimous interior profile arises only from its all-aligned truth and has posterior u.

For a tie, take a labeled half-set B and a non-prior truth u. The profile with B reporting u and B^c reporting $\rho(u)$ has two equal-mass sources: truth u with B aligned and truth $\rho(u)$ with B^c aligned. Their type probabilities are equal and symmetry gives $\tau(u) = \tau(\rho(u))$. The posterior mean is therefore $(u + \rho(u))/2 = \mu_0$. A prior tie collapses to a unanimous prior profile and also has posterior μ_0. The margin rule selects the posterior mean in all cases.

Different margins cannot collide on path: the multiplicities of the two reported values identify c, ℓ, and hence d; the labeled majority coordinates identify B; and interior tags were chosen separately. Thus the preceding posterior calculations do not omit cross-channel sources.

Step 6: establish the saddle inequalities. By Step 5 and Lemma 1, the margin rule is a Bayes response to β^*. By its construction, every truthful-majority type profile leaves the truthful majority at its smallest margin and achieves clipping loss $(m - t_d)_+^2$; every fake-majority type profile sends a common opposite-side exterior report and reaches the far boundary, achieving $(m + t_d)^2$; and every tie uses reflection and reaches loss m^2. Hence β^* attains the conditional suprema in Proposition 1 state by state and type profile by type profile.

Let a^* be the margin rule and let $L(a, \beta)$ denote expected quadratic loss. The two best response properties give, for all a and β,

$$L(a^*, \beta) \leq L(a^*, \beta^*) \leq L(a, \beta^*).$$

The left inequality says β^* is worst for a^*; the right says a^* is Bayes-optimal against β^*. Thus (a^*, β^*) is a saddle point. Subtracting its loss from the prior loss and using the event decomposition in Appendix A.4 yields (8). This proves Theorem 2 for arbitrary K and for every rule class allowed by the model.

A.6 The two-adviser algebra

For $K = 2$, the only strict-majority margins are $d = 2$. There are three type-count events: no misaligned advisers, a tie with one misaligned adviser, and two misaligned advisers. The tie is value-neutral. Therefore Theorem 2 gives

$$V_2 = \alpha^2 G(t_2) - (1 - \alpha)^2 D(t_2),$$

which is (9). Put $A = \alpha^2$, $B = (1 - \alpha)^2$, and $t = t_2$. The radius equation is

$$A\mathbb{E}[(\widetilde{m} - t)_+] = B(t + M). \tag{25}$$

Since

$$\widetilde{m}^2 - (\widetilde{m} - t)_+^{;2} = \min\{\widetilde{m}, t\}^2 + 2t\,(\widetilde{m} - t)_+\,,$$

we have

$$\begin{aligned} AG(t) - BD(t) &= A\mathbb{E}[\min\{\widetilde{m}, t\}^2] + 2At\mathbb{E}[(\widetilde{m} - t)_+] - B(2Mt + t^2) \\ &= A\mathbb{E}[\min\{\widetilde{m}, t\}^2] + 2Bt(t + M) - B(2Mt + t^2) \\ &= A\mathbb{E}[\min\{\widetilde{m}, t\}^2] + Bt^2, \end{aligned}$$

where the second line uses (25). This proves (10).

A.7 Redundancy asymptotics

Proof of Corollary 2(i) and (iii). Let $p_0 = \mathbb{P}(\widetilde{m} > 0)$ and let $m_{\min}$ be the smallest positive value of $\widetilde{m}$. For $0 \le t < m_{\min}$,

$$\mathbb{E}[(\widetilde{m} - t)_+] = M - p_0 t, \tag{26}$$

$$G(t) = 2Mt - p_0 t^2, \qquad D(t) = 2Mt + t^2. \tag{27}$$

Pair the event with $j < K/2$ misaligned advisers and its complementary event with $K - j > K/2$ misaligned advisers. Put $d = K - 2j$ and

$$A = \binom{K}{j}\alpha^{K-j}(1-\alpha)^j, qquadB = \binom{K}{j}\alpha^j(1-\alpha)^{K-j}.$$

Their contribution to V_K is $AG(t_d) - BD(t_d)$. When α is sufficiently close to one half, $t_d < m_{\min}$, and the radius first-order condition, using (26), is

$$A(M - p_0 t_d) = B(M + t_d).$$

Solving gives

$$t_d = \frac{M(A - B)}{Ap_0 + B}. \tag{28}$$

Substituting (28) into (27) yields the exact local-channel value

$$AG(t_d) - BD(t_d) = \frac{M^2(A - B)^2}{Ap_0 + B}. \tag{29}$$

Write $\alpha = 1/2 + \varepsilon$. For fixed K, j,

$$A - B = \binom{K}{j} 2^{-K}(4d\varepsilon) + O(\varepsilon^2),$$
$$Ap_0 + B = \binom{K}{j} 2^{-K}(1 + p_0) + O(\varepsilon).$$

Hence the margin-d channel contributes

$$\frac{16M^2}{1+p_0}\binom{K}{j} 2^{-K} d^2\varepsilon^2 + o(\varepsilon^2).$$

If $X_1, \ldots, X_K$ are independent random signs, then

$$\sum_{j=0}^{K}\binom{K}{j}(K-2j)^2 = 2^K \mathbb{E}\left[\left(\sum_{i=1}^{K} X_i\right)^2\right] = K2^K.$$

Positive and negative margins make equal contributions and a zero margin contributes zero, so

$$\sum_{j<K/2}\binom{K}{j}(K-2j)^2 = K2^{K-1}.$$

Summing (29) over positive margins gives

$$V_K(1/2+\varepsilon) = \frac{8KM^2}{1+p_0}\varepsilon^2 + o(\varepsilon^2).$$

The ratio to the $K = 1$ expression converges to K, proving (i).

For (iii), write $f(1/2 + \varepsilon) = 1/2 + x_\varepsilon$. Local monotonicity of V_1 above the threshold selects a unique small positive branch. The equality $V_1(f) = V_2(1/2 + \varepsilon)$ and the preceding expansion imply

$$Cx_\varepsilon^2 + o(x_\varepsilon^2) = 2C\varepsilon^2 + o(\varepsilon^2), \qquad C = \frac{8M^2}{1+p_0} > 0.$$

Taking the positive square root yields $x_\varepsilon = \sqrt{2}\varepsilon + o(\varepsilon)$. □

Proof of Corollary 2(ii). Let $R_K = \mathrm{Var}(\mu) - V_K$ be the worst-case loss of the optimal rule. Let $J \sim \mathrm{Binomial}(K, 1-\alpha)$ count misaligned advisers. On $J = K/2$ the conditional loss is $\mathrm{Var}(\mu)$, and on $J > K/2$ it is $\mathbb{E}[(\widetilde{m} + t_{2J-K})^2] \geq \mathrm{Var}(\mu)$. Therefore

$$R_K \geq \mathrm{Var}(\mu)\mathbb{P}(J \geq K/2). \tag{30}$$

The standard binomial large-deviation limit gives

$$-\lim_{K\to\infty}\frac{1}{K}\log\mathbb{P}(J\geq K/2)=\mathrm{KL}\left(\frac{1}{2}\middle\|1-\alpha\right)=\mathrm{KL}\left(\frac{1}{2}\middle\|\alpha\right). \tag{31}$$

For the matching upper bound, split R_K into bad-or-tied and good-majority events. Since $0\leq\widetilde{m},t_d\leq 1$, every bad-or-tied conditional loss is at most 4, so that part is at most $4\mathbb{P}(J\geq K/2)$.

For a good-majority margin $d=K-2j>0$, finite support implies that $p_{\max}:=\mathbb{P}(\widetilde{m}=\bar{m})>0$. The radius equation gives

$$p_{\max}(\bar{m}-t_d)\leq\mathbb{E}[(\widetilde{m}-t_d)_+]=r^{-d}(t_d+M)\leq(\bar{m}+M)r^{-d}.$$

Consequently

$$\mathbb{E}[(\widetilde{m}-t_d)_+^2]\leq Cr^{-2d} \tag{32}$$

for a constant C independent of d and K. The good-majority loss is bounded by

$$C\sum_{j<K/2}\binom{K}{j}\alpha^{K-j}(1-\alpha)^j r^{-2(K-2j)}.$$

For $x=j/K\in[0,1/2]$, the exponential rate of a summand is

$$H(x)=\mathrm{KL}(x\,\|\,1-\alpha)+2(1-2x)\log r.$$

This convex function is decreasing on $[0,1/2]$: its derivative is increasing and

$$H'(1/2)=\log r-4\log r=-3\log r<0.$$

Hence its minimum on the interval is at $x=1/2$, where $H(1/2)=\mathrm{KL}(1/2\,\|\,\alpha)$. Standard type-count bounds (there are only $K+1$ binomial types) therefore show that the good-majority sum has exponential rate no smaller than this value. Combined with the bad-or-tied bound, this implies

$$\liminf_{K\to\infty}-\frac{1}{K}\log R_K\geq\mathrm{KL}\left(\frac{1}{2}\middle\|\alpha\right).$$

The lower bound (30) and (31) give the reverse inequality for the limit superior. Dividing R_K by the positive constant $\mathrm{Var}(\mu)$ does not change the exponent, which proves (ii). □

B Numerical audit details

The replication code uses the symmetric support $S=\{0,1/8,\ldots,1\}$ with uniform probabilities. For each $\alpha\in\{.55,.65,.80,.95\}$ and $K=1,\ldots,6$, it performs the following deterministic checks.

1. It solves (6) by bisection and evaluates (8).

2. It builds the labeled product transports (23), enumerates every state and type mask, and records the joint mass of every on-path report profile. It verifies that conditional rows sum to one, total probability sums to one, the margin action equals the posterior mean, the constructed conditional loss equals (18), and posterior variance equals the formula value.

3. For $K \leq 3$, it independently discretizes the decision-maker's actions and solves an epigraph linear program over all report profiles. The grid includes every support belief, prior, and trust boundary, so the analytical optimum is in the feasible action set.

The largest conditional probability-row discrepancy across the 24 saddle audits is below 9×10^{-13}; the largest posterior-action or pointwise-loss discrepancy is below 2.3×10^{-16}; and the largest aggregate formula discrepancy is below 1.1×10^{-14}. The largest formula–LP discrepancy is 1.4×10^{-16}. Exact unrounded results and every radius are stored in the JSON output. No random seed is required because no random sampling is used.

References

Alonso, R., and G. Padró i Miquel (2025): "Competitive Capture of Public Opinion," *Econometrica*, 93, 1265–1297.

Alonso, R., T. Gan, and J. Hu (2026): "Robust Delegation," CEPR Discussion Paper 21382.

Ambrus, A., and S. Takahashi (2008): "Multi-Sender Cheap Talk with Restricted State Spaces," *Theoretical Economics*, 3, 1–27.

Battaglini, M. (2002): "Multiple Referrals and Multidimensional Cheap Talk," *Econometrica*, 70, 1379–1401.

Blackwell, D. (1951): "Comparison of Experiments," in *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, 93–102.

Crawford, V. P., and J. Sobel (1982): "Strategic Information Transmission," *Econometrica*, 50, 1431–1451.

Dworczak, P., and A. Smolin (2026): "Robust Trust," arXiv:2602.09490, https://arxiv.org/abs/2602.09490.

Fudenberg, D., and A. Liang (2025): "Friend or Foe: Delegating to an AI Whose Alignment is Unknown," arXiv:2509.14396, https://arxiv.org/abs/2509.14396.

Irving, G., P. Christiano, and D. Amodei (2018): "AI Safety via Debate," arXiv:1805.00899.

Khan, A., J. Hughes, D. Valentine, L. Ruis, K. Sachan, A. Radhakrishnan, E. Grefenstette, S. R. Bowman, T. Rocktäschel, and E. Perez (2024): "Debating with More Persuasive LLMs Leads to More Truthful Answers," *Proceedings of ICML*.

Krishna, V., and J. Morgan (2001): "A Model of Expertise," *Quarterly Journal of Economics*, 116, 747–775.